

Table 3: Hyperparameter sensitivity on UDIAT-B (Dice, %). Default settings are **bolded**; $\uparrow$ marks peak performance.

Temp. τ	Dice	Scales S	Dice	λ_{ctx}	Dice	N (LPE)	Dice	Prob. α	Dice
0.05	84.8	1	79.8	0.1	84.5	2	84.6	0.05	85.2
0.10	85.9$\uparrow$	2	83.0	0.5	85.3	4	85.9$\uparrow$	0.10	85.6
0.20	85.2	3	84.6	1.0	85.9$\uparrow$	6	85.7	0.15	85.9$\uparrow$
0.50	84.1	4	85.9$\uparrow$	2.0	85.0	8	85.5	0.20	85.4

A.1 Supplementary Experimental Details

Data augmentation. Both views follow the BYOL protocol: random crop (0.2–1.0), horizontal flip, colour jitter, Gaussian blur, and grayscale; $\mathbf{v}_2$ additionally applies solarisation. Geometric transforms are applied identically to masks.

Downstream fine-tuning. For **classification** (POCUS, BUSI), only the last three backbone layers are fine-tuned using Adam (lr = 0.01) for 30 epochs. For **segmentation** (UDIAT-B, TN3K, BUSI, HMC-QU), we employ Mask-RCNN with the pre-trained backbone, optimised via SGD (weight decay 10^{-4}).

Mask sampling. Each binary mask is downsampled via adaptive average pooling to match backbone feature maps at $\{7 \times 7, 14 \times 14, 28 \times 28, 56 \times 56\}$. Weighted by spatial coverage, K masks are drawn from the resulting categorical distribution, ensuring anatomically salient structures receive sampling priority.

Computational cost. LP-SAM adapter fine-tuning introduces **13M** parameters and requires ~ 24 h; the latent prompt engine adds **11.1M** parameters (~ 8 h). Full ANAUS pre-training (300 epochs, batch 192) completes in ~ 10 h on two RTX 4090 GPUs (~ 22 GB VRAM). Both LP-SAM stages are *one-time* procedures excluded from SSL iterations. The retained encoder processes a 224×224 image in $\sim \mathbf{0.45}$ ms on a RTX 4090, confirming *real-time clinical viability*.

A.2 Hyperparameter Sensitivity

Table 3 reports UDIAT-B Dice under systematic variation of key hyperparameters. $\tau=0.1$ sharpens *inter-anatomy separability* without over-penalising hard negatives; $\lambda_{\text{ctx}}=1.0$ and $N=4$ cross-attention rounds yield stable peaks, with diminishing returns beyond these values. For the Stage 1 adapter loss, the combined formulation $(1-\alpha)\mathcal{L}_{\text{BCE}} + \alpha\mathcal{L}_{\text{Dice}}$ improves mask IoU, confirming that the Dice term is critical for foreground–background imbalance in US images.

A.3 Pre-training Resolution, Data Volume, and K

Fig. 5 presents three complementary analyses. (a) $K=16$ achieves the best PPV/Sensitivity/Dice balance; $K=32$ improves PPV but degrades Sensitivity, suggesting *redundant masks introduce noise*. (b) Resolution 224×224 preserves fine-grained anatomical detail essential for boundary delineation; (c) The

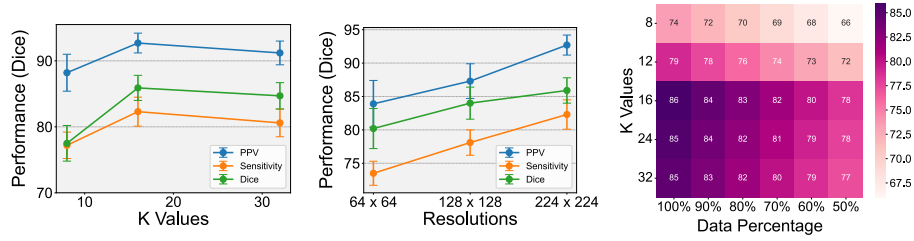


Fig. 5: Ablation on UDIAT-B. (a) Effect of mask sample count K . (b) Pre-training resolution. (c) Joint effect of K and data percentage on Dice.

Table 4: ANAUS vs. IVPP++ across backbone scales on UDIAT-B.

Backbone	Method	PPV $\uparrow$	Sens. $\uparrow$	Dice $\uparrow$
ResNet-18 (0.45 ms)	IVPP++	89.1	80.3	83.6
	ANAUS	92.7	82.3	85.9
ResNet-50 (1.12 ms)	IVPP++	90.2	81.1	84.5
	ANAUS	93.5	83.0	86.5
ResNet-101 (1.84 ms)	IVPP++	90.8	81.7	85.1
	ANAUS	94.1	83.5	87.3
ResNet-152 (2.53 ms)	IVPP++	91.3	82.0	85.4
	ANAUS	94.6	84.2	87.5

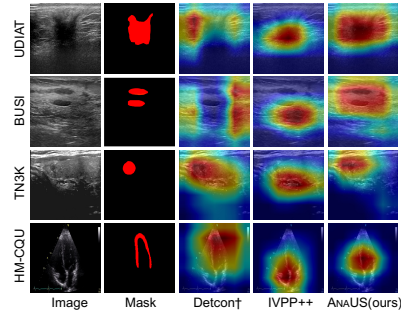


Fig. 6: Grad-CAM visualization. ANAUS focuses on anatomy with minimal background dispersion.

heatmap confirms the joint optimum at $K=16$ with 100% data; performance degrades notably below 60% data under suboptimal K , underscoring the complementary roles of data volume and mask sampling.

A.4 Impact of Backbone Scale

Table 4 compares ANAUS and IVPP++ across four ResNet variants on UDIAT-B. ANAUS consistently outperforms at every scale, with the Dice gap sustained from +2.3% (ResNet-18) to +2.1% (ResNet-152). Notably, ANAUS with ResNet-18 already surpasses IVPP++ with ResNet-50, demonstrating that anatomy-anchored pre-training compensates for limited model capacity.

A.5 Grad-CAM Interpretability

Fig. 6 visualizes final-layer activations via Grad-CAM across four datasets. DetCon $\dagger$ produces diffuse activations extending beyond lesion boundaries; IVPP++ improves localization but retains non-negligible background response. ANAUS generates activations *tightly confined to target anatomy*, confirming that the anatomy-level contrastive objective suppresses background encoding and internalises diagnostically relevant spatial priors and representative features.